Lec 5. Families of Distributions

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Learning objectives

After this section, you should be able to ...

- understand the definition of exponential family and its properties
- list common distributions in exponential family
- know the location-scale distribution family

5.1 Exponential Family

Exponential family comprises a set of flexible distribution ranging both continuous and discrete random variables. The members of this family have many important properties which merits discussing them in some general format. Many of the probability distributions are specific members of this family:

- Gaussian: $\mathbb{R}^p$
- Multinomial: categorical
- Binomial: counts of success/failure
- Poisson: $\mathbb{N}^+$
- Laplace: $\mathbb{R}^+$
- Exponential: $\mathbb{R}^+$
- Beta: (0,1)
- Weibull: $\mathbb{R}^+$
- Weishart: symmetric positive-definite matrices

Definition

A family $\{F_\theta, \theta \in \Theta\}$ of distributions forms an (k-dimensional) exponential family if the distributions F_θ have densities (pdf/pmf) of the form:

$$f(x, \theta) = \exp\left[\sum_{i=1}^{k} Q_i(\theta) T_i(x) - A(\theta)\right] h(x) \tag{5.1}$$

where A, Q_i are functions which map θ to $\mathbb{R}$; $h(x) > 0$ and $T_i(x)$ are statistics.

- The support of exponential family,

$$\mathcal{G} = \{x : f(x, \theta) > 0\} = \{x : h(x) > 0\}$$

  is **irrelevant with the parameter θ**. i.e., exponential family has common support.

- The exponential family structure is preserved for an i.i.d. sample

The One-Parameter Case

Consider a sample: $X_1, \ldots, X_n$ *iid* $\sim f$, where $f \in \mathcal{F} = \{f(x, \theta), \theta \in \Theta \subset \mathbb{R}\}$ is a one-parameter exponential family distribution with density/pmf function

$$f(x, \theta) = h(x) \exp\{Q(\theta)T(x) - A(\theta)\} \tag{5.2}$$

The sample $\mathbf{X} = (X_1, \ldots, X_n)$ is a random vector with density/pmf:

$$\begin{aligned} f(\mathbf{x}|\theta) &= \prod_{i=1}^{n} \left(h\left(x_i\right) \exp\left[Q(\theta)T\left(x_i\right) - A(\theta)\right]\right) \\ &= \left[\prod_{i=1}^{n} h\left(x_i\right)\right] \times \exp\left[Q(\theta) \sum_{i=1}^{n} T\left(x_i\right) - nA(\theta)\right] \\ &= h^*(x) \exp\left\{Q^*(\theta)T^*(\mathbf{x}) - B^*(\theta)\right\} \end{aligned}$$

where: $h^*(x) = \prod_{i=1}^{n} h\left(x_i\right), Q^*(\theta) = Q(\theta), T^*(\mathbf{x}) = \sum_{i=1}^{n} T\left(x_i\right), B^*(\theta) = nA(\theta)$.

5.1.1 Canonical Exponential Family

The One-Parameter Case

- Re-parametrize setting $\eta = Q(\theta)$ in (5.2)—the Natural Parameter
- The density has the form

$$f(x, \eta) = h(x) \exp\{\eta T(x) - A(\eta)\}$$

- The function $A(\eta)$ is defined as the log-normalization constant:

$$A(\eta) = log \int_{\mathcal{X}} h(x) \exp\{\eta T(x)] dx$$

  if X is continuous or

$$A(\eta) = log \sum_{x \in \mathcal{X}} h(x) \exp\{\eta T(x)]$$

  if X is discrete

- The Natural Parameter Space $\mathcal{E} = \{\eta : \exp\{A(\eta)\} < \infty\}$ is a convex set.

The One-Parameter Case

Theorem 1. *Suppose X is distributed according to a canonical exponential family, i.e., the density/pmf function is given by*

$$f(x, \eta) = h(x) \exp[\eta T(x) - A(\eta)], \text{ for } x \in \mathcal{X}$$

If η_0 is an interior point of $\mathcal{E}$, the natural parameter space, then

- *The* ***moment generating function*** *of $T(X)$ exists under η_0 and is given by*

$$M_T(s) = E_{\eta_0}\left[e^{sT(X)}\right] = exp(A(s + \eta_0) - A(\eta_0))$$

  for all s such that $\eta_0 + s \in \mathcal{E}$.

- *The function* $e^{A(\eta)}$ *is infinitely differentiable inside the integral(sum) sign at every* η_0

- $E_{\eta_0}[T(X)] = A'(\eta_0)$

- $\mathrm{Var}_{\eta_0}[T(X)] = A''(\eta_0)$

The Multi-parameter Case

- Let the model (5.1) be indexed by $\eta = (Q_1, \ldots, Q_k)^{\mathrm{T}}$ rather than θ. Thus, the canonical k-parameter exponential family generated by $\mathbf{T}$ and h is

$$f(x, \boldsymbol{\eta}) = h(x) \exp\left\{\eta^{\mathrm{T}} \mathbf{T}(x) - A(\boldsymbol{\eta})\right\}, x \in \mathcal{X} \tag{5.3}$$

  where $\mathbf{T}(x) = (T_1(x), \ldots, T_k(x))^{\mathrm{T}}$ and, in the continuous case,

$$A(\eta) = log \int_{\mathcal{X}} h(x) \exp\left\{\eta^{\mathrm{T}} \mathbf{T}(x)\right\} dx$$

  In the discrete case, $A(\eta)$ is defined in the same way except integrals over $\mathcal{X}$ are replaced by sums.

- The natural parameter space

$$\mathcal{E} = \{\boldsymbol{\eta} : \exp\{A(\boldsymbol{\eta})\} < \infty\}$$

  is a convex set and $A(\boldsymbol{\eta})$ is a convex function.

- **Minimal representations:** An exponential family representation is said to be minimal if the sufficient statistics are not redundant, i.e. there is no set of coefficients $a \in \mathbb{R}^k, a \neq 0$ such that,

$$\sum_{i=1}^{k} a_i T_i(x) = \text{ const}$$

  for all $x \in \mathcal{X}$. Non-minimal exponential families are not statistically identifiable (while minimal ones are).

- **full-rank family**: $dim\mathcal{E} = k$.

- **curved exponential family**: $dim\mathcal{E} < k$, e.g., $N(\mu, \mu^2), N_2(0, 0, 1, 1, \rho)$
  .

- At any interior point $\boldsymbol{\eta}$, $e^{A(\boldsymbol{\eta})}$ is infinitely partially differentiable with respect to each η_i, and the partial derivatives of any order can be obtained by differentiating inside the integral sign.

- $E_\eta\left[T_i(X)\right] = \frac{\partial}{\partial \eta_i} A(\boldsymbol{\eta})$;
  $\mathrm{Cov}_\eta\left(T_i(X), T_j(X)\right) = \frac{\partial^2}{\partial \eta_i \partial \eta_j} A(\boldsymbol{\eta}), 1 \leq i, j \leq k.$

- If $\boldsymbol{\eta}, \boldsymbol{s}$ are such that $\boldsymbol{\eta}, \boldsymbol{\eta} + \boldsymbol{s}$ are in the interior of $\mathcal{E}$, then the joint mgf of $(T_1(X), \cdots, T_k(X))$ exists and equals

$$M_\eta(\boldsymbol{s}) = e^{A(\boldsymbol{\eta}+\boldsymbol{s}) - A(\boldsymbol{\eta})}$$

The Multi-parameter Case

Theorem 2. *If the distribution of X belongs to a canonical exponential family (5.3) and η is an interior point of $\mathcal{E}$, then for any bounded integrable function $g(x)$, the following function*

$$G(\eta) = \int \cdots \int g(x) \exp\left\{\eta^{\mathrm{T}} \mathbf{T}(x)\right\} h(x) dx$$

satisfy

$$\begin{aligned}
\frac{\partial^m G(\eta)}{\partial \eta_1^{m_1} \cdots \partial \eta_k^{m_k}} &= \int \cdots \int \frac{\partial^m}{\partial \eta_1^{m_1} \cdots \partial \eta_k^{m_k}} \left[g(x) \exp\left\{\eta^{\mathrm{T}} \mathbf{T}(x)\right\} h(x)\right] dx \\
&= \int \cdots \int T_1^{m_1} \cdots T_k^{m_k} \left[g(x) \exp\left\{\eta^{\mathrm{T}} \mathbf{T}(x)\right\} h(x)\right] dx
\end{aligned}$$

where $m_1 + \cdots + m_k = m$.

Sufficiency and completeness

Theorem 3. *If a sample X is distributed according to a full-rank exponential family, then $T(X) = (T_1(X), \ldots, T_k(X))$ is minimal sufficient.*

In an exponential family, it turns out that not only is the statistic T minimal sufficient, it is complete.

Theorem 4. *If a sample X is distributed according to a full-rank exponential family with minimal sufficient statistic $T(X) = (T_1(X), \ldots, T_k(X))$, and the natural parameter space contains open subset of $\mathbb{R}^k$, then T is complete.*

5.1.2 The maximum entropy duality

- Suppose that we are given a random sample $\{X_1, \ldots, X_n\}$ from some distribution, and we compute the empirical expectations of certain functions that we choose:

$$\widehat{\mu}_i = \frac{1}{n} \sum_{j=1}^{n} T_i\left(X_j\right) \qquad \text{for} \qquad i \in \{1, \ldots, k\}$$

- Based on just these empirical expectations we want to infer a full probability distribution on the samples. A distribution p is consistent with the data we observe if it is the case that,

$$\widehat{\mu}_i = \mathbb{E}_p\left[T_i(X)\right] \qquad \text{for } i \in \{1, \ldots, k\}$$

- The entropy of a distribution is:

$$H(p) = -\int p(x) \log(p(x)) dx$$

- The principle of maximum entropy suggests to pick the distribution that has the largest (Shannon) entropy. So we could imagine trying to find the distribution p^* that,

$$p^* = \arg\max_p H(p)$$

subject to the constraints that,

$$\widehat{\mu}_i = \mathbb{E}_p\left[T_i(X)\right] \qquad \text{for } i \in \{1, \ldots, k\}.$$

The solution to this problem can shown to have the form

$$p^*(x) = \exp\left[\sum_{i=1}^{k} \theta_i T_i(x) - A(\theta)\right] h(x).$$

This provides a motivation for exponential families.

Some Distributions not in the Exponential Family

- Irregular distributions whose support depends on the parameter cannot be members of the Exponential family, e.g., $U(0, \theta)$, $U(-\theta, \theta)$, $f(x|\theta) = e^{\theta - x} I(x > \theta)$

- Cauchy distributions $f(x \mid \mu) = \frac{1}{\pi\left[1+(x-\mu)^2\right]}, x \in \mathbb{R}$

5.2 Location and Scale Families

Definition

Let $f(x)$ be any pdf. Then for any μ, $-\infty < \mu < \infty$, and any $\sigma > 0$, the family of pdfs

$$\frac{1}{\sigma} f(\frac{x-\mu}{\sigma}),$$

indexed by the parameter (μ, σ), is called the location-scale family with standard pdf $f(x)$; μ is called the location parameter and σ is called the scale parameter.

if $Z \sim f(z)$, $X \sim 1/\sigma f((x-\mu)/\sigma)$,

- $X \stackrel{d}{=} \sigma Z + \mu$
- $EX = \sigma EZ + \mu, Var(X) = \sigma^2 Var(Z)$
- location family: $\{f(x-\mu), \mu \in \mathbb{R}\}$
- scale family: $\{f(x/\sigma)/\sigma, \sigma > 0\}$

Example

(Location family ancillary statistic) Let $X_1, \ldots, X_n$ be iid observations from a location parameter family with cdf $F(x-\theta), -\infty < \theta < \infty$. Then $R = X_{(n)} - X_{(1)}$, is an ancillary statistic.

Example

(Scale family ancillary statistic) Let $X_1, \ldots, X_n$ be iid observations from a scale parameter family with cdf $F(x/\sigma), \sigma > 0$. Then $X_1/X_n, \ldots, X_{n-1}/X_n$ is an ancillary statistic.

Example

If $X_1, \ldots, X_n$ is a random sample from a location-scale family with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, then, for any $(i, j, k), 1 \leq i, j, k \leq n$, $(X_i - X_k) / (X_j - X_k)$ and $\left(X_{(i)} - X_{(k)}\right) / \left(X_{(j)} - X_{(k)}\right)$ are ancillary.